CAP-01 / DOCUMENT 03

Requirements Gathering and Traceability

Shopping Copilot

Arabic-first shopping advice and guarded Storefront actions

Programme	Digital Egypt Pioneers Initiative (DEPI)
Track	AWS ML Engineering
Submission deadline	16 October 2026
Export date	16 October 2026
Source document date	16 October 2026
Release status	Prepared for submission; official upload pending

Prepared for the AWS ML Engineering graduation submission. The owner confirms lecturer approval and the organization-repository invitation were received. Approval wording/date and numerical grading details were not supplied. Assigned responsibilities are distinguished from actual contributions.

Requirements gathering and traceability

Project: Shopping Copilot. **Updated:** 16 October 2026. **Track:** AWS ML Engineering. **Published DEPI deadline:** 16 October 2026. **Status:** prepared for submission under the supplied content guidelines. Lecturer approval and the repository invitation are owner-confirmed; a dated approval reference and official upload record have not been supplied.

Actual working arrangement (2 October 2026): the owner and coding assistant perform the work; the owner coordinates this deliverable. Earlier named academic leads were administrative allocations, not verified contributions or available staffing. See the updated CAP-01 register.

Problem and stakeholders

A Shopper may know what they want but struggle to choose among products or translate a request into filters, navigation and cart controls. The project addresses both decision assistance and execution on one Controlled Storefront. It must support natural multilingual input while preserving the Shopper's control over consequential actions.

Stakeholders are the Shopper, the project team maintaining the catalogue/application, and the DEPI reviewers assessing the graduation deliverables. Team technical roles and confirmed dates are recorded in the CAP-01 register. The deployment is a controlled graduation demonstration with fictional commerce data, not a service connected to real retailers or payment providers.

Elicitation sources and their limits

Source: Owner-approved local specification and domain model

What it establishes: Supported journeys, controlled scope and safety invariants

Limitation: Earlier wording is superseded where later owner decisions are explicit.

Source: Informal feedback from five testers, relayed by the owner

What it establishes: Positive reactions to advice; two successful account navigations; demand for natural styling and comparisons

Limitation: No standardized scores, complete task matrix or measured comparative benefit.

Source: Owner observations and approved changes

What it establishes: Need for exact-match explanations, conversational follow-up and model-owned interpretation

Limitation: Individual reports are not population-level research.

Source: Runtime tests and live reports

What it establishes: Concrete observed successes, failure modes and guarded behaviour

Limitation: Coverage is finite and tied to recorded configurations.

Source: Owner-supplied DEPI schedule and team table

What it establishes: Applicable dates and assigned technical roles

Limitation: The images do not establish official rubric weights or a document template.

Source: Five supplied DEPI PDFs, reviewed 2 October

What it establishes: Required documentation content, infographic, organization GitHub delivery and confirmed dates.

Limitation: Literature/lecturer heading discrepancy; grading weights and actual repository access not supplied.

All five testers reportedly lacked confidence choosing purchases online; this is not evidence that all had difficulty operating website controls. The recorded findings were robotic responses and insufficient comparison/decision support. UI comments were not supplied in detail and are not invented as requirements. The approved informal study and technical release exceptions are fully recorded in Ticket 16 closeout.

Functional requirements

User stories and use-case mapping

These stories express requirements rather than quotations from participants. Detailed preconditions, alternative flows and observable outcomes are in the design use cases.

Story: US-01

Shopper goal: As a Shopper, I want to express budget, size and exclusions in my own language so I can find suitable products.

Requirement / use case: FR-01-02 / UC-01

Acceptance example: Preserve explicit exclusions; unknown facts cannot qualify a product as an Exact Match.

Story: US-02

Shopper goal: As a Shopper, I want to discuss trade-offs so I can make a more informed choice.

Requirement / use case: FR-03 / UC-02

Acceptance example: Compare verified facts, label styling opinions, acknowledge unknown qualities; no mutation.

Story: US-03

Shopper goal: As a Shopper, I want to open an identified product or find a control so I can continue shopping.

Requirement / use case: FR-04 / UC-03

Acceptance example: Navigate only to supported observed destinations; Spotlight does not submit checkout.

Story: US-04

Shopper goal: As a Shopper, I want to edit one cart line and undo a mistake so other items stay unchanged.

Requirement / use case: FR-05 / UC-04-05

Acceptance example: Resolve the exact variant; distinguish added quantity from total; preserve the original Undo expiry.

Story: US-05

Shopper goal: As a Shopper, I want to approve the precise effect before clearing the cart or submitting fictional checkout.

Requirement / use case: FR-06 / UC-06

Acceptance example: Changed cart state or expired/reused Confirmation cannot authorize the effect.

Story: US-06

Shopper goal: As a Shopper, I want account/order guidance while entering private details myself.

Requirement / use case: FR-07 / UC-07

Acceptance example: Sensitive values are excluded from observations and model requests.

Story: US-07

Shopper goal: As a Shopper, I want to stop or refresh without accidentally repeating a cart operation.

Requirement / use case: FR-08 / UC-08

Acceptance example: Reject stale results and reconcile uncertain outcomes before any further effect.

Story: US-08

Shopper goal: As a Shopper, I want to review dictated text before sending it and type when speech is unavailable.

Requirement / use case: FR-09 / UC-09

Acceptance example: Transcription is editable; recording/availability failures do not block typing.

Functional acceptance

Priority **Must** means required within the approved graduation scope; it does not imply every future cloud or ML requirement is already implemented.

ID: FR-01

Requirement: Interpret Egyptian Arabic, Franco-Arabic, English and mixed-language shopping requests, including corrections and references.

Observable acceptance and evidence: Evaluate labelled intent/Constraints separately from completed browser tasks; report failures by language/scenario.

State: Local implementation/evidence exists; new dataset evaluation planned.

ID: FR-02

Requirement: Discover available products using explicit requirements, budget, size, colour and ordering preferences.

Observable acceptance and evidence: Verify catalogue eligibility; distinguish Exact Matches from Alternatives with named unmet or unknown requirements.

State: Implemented and tested locally.

ID: FR-03

Requirement: Provide natural advice, relevant comparisons and concise reasons for exact matches using verified catalogue facts.

Observable acceptance and evidence: Opinions are distinguishable from facts; absent qualities remain unknown; ask a useful question when needed without forcing a questionnaire.

State: Implemented; qualitative evidence and known grounding limitations recorded.

ID: FR-04

Requirement: Navigate or Spotlight the requested Storefront destination and open an identified product.

Observable acceptance and evidence: Use current observed targets and same-origin destinations; do not confuse locating a control with activating it.

State: Implemented and tested locally.

ID: FR-05

Requirement: Add a selected available variant, change quantity, remove a line and offer ten-second Undo.

Observable acceptance and evidence: Assert exact variant/quantity and unaffected lines; Undo restores the precise prior state while its original deadline remains valid.

State: Implemented; automated and recorded live checks.

ID: FR-06

Requirement: Require fresh, bound, single-use Confirmation for bulk clearing and checkout submission.

Observable acceptance and evidence: Unconfirmed, expired, mismatched or reused authorization cannot mutate state.

State: Implemented; current regressions and historical live evidence.

ID: FR-07

Requirement: Support account/order guidance and newest-order Spotlight without handling sensitive login/payment fields.

Observable acceptance and evidence: Shopper enters sensitive values directly; agent acts only on permitted controls.

State: Implemented; two reported account navigations are a narrower human observation.

ID: FR-08

Requirement: Support Stop, refresh reconciliation and task ownership across tabs.

Observable acceptance and evidence: Discard late/stale Actions and do not replay an uncertain mutation automatically.

State: Implemented and tested locally.

ID: FR-09

Requirement: Offer readable chat, RTL/localization, product cards and editable speech input when the selected service is available.

Observable acceptance and evidence: The Shopper can type when speech is unavailable and can review transcription before sending.

State: Implemented; paid voice currently constrained by provider-key allowance.

All functional requirements above are Must within supported local capabilities. They do not extend the catalogue to laptops merely because a participant used laptops as a comparison example.

Safety and quality requirements

ID: SQ-01

Requirement: The LLM owns interpretation; validation must not become a shopper-language keyword/regex interpreter.

Verification approach: Inspect boundaries and test schema/target/quantity failures independently from live language-quality cases.

ID: SQ-02

Requirement: Reject off-origin navigation and exclude Sensitive Field values from model context.

Verification approach: Adversarial browser and deterministic policy regressions.

ID: SQ-03

Requirement: Validate inventory, quantity/price bounds, observed targets and exact mutation authorization.

Verification approach: Public API, Bridge and Storefront regression checks, including stale and duplicate attempts.

ID: SQ-04

Requirement: Preserve truthful evidence and safe fallback.

Verification approach: Retain failures, unknown usage and original artifacts; malformed advice preserves verified cards without retry loops or mutation.

ID: SQ-05

Requirement: Measure latency, task steps and cost with explicit denominators.

Verification approach: Separate model-inclusive time from browser execution; count repairs and do not substitute estimates for missing billing.

ID: SQ-06

Requirement: Prevent credential and participant-data disclosure.

Verification approach: Keep environment files/secrets out of Git, omit personal contacts from public reports, and obtain authorization for recording/data reuse.

ID: SQ-07

Requirement: Keep advice read-only and resistant to untrusted instructions in product text.

Verification approach: No executable fields/tools in advice responses; later action requests re-enter normal execution guards.

Current evidence is summarized in RESULTS.md. The historical 44-case full-model gate and its performance definitions remain tied to its frozen runtime. The local release uses the owner's explicit evidence exception; it does not prove every paraphrase, product claim or browser interaction is correct.

Graduation ML and deployment requirements

ID: GR-01

Requirement: Release reviewed multilingual data with provenance and exposure history.

Planned acceptance: CAP-02: documented labels, reviewer decisions, group-aware splits and an untouched evaluation set.

ID: GR-02

Requirement: Train an offline classical baseline and compare it fairly with the LLM.

Planned acceptance: CAP-03: TF-IDF + Logistic Regression, train-only learned transforms, validation-based selection, same eligible inputs, confusion matrices and error/cost/latency analysis.

ID: GR-03

Requirement: Deploy the Controlled Storefront application on AWS.

Planned acceptance: CAP-04-07: reviewed budget/design, qualified model, protected deployment, operational acceptance and repeatable recovery/rollback evidence.

ID: GR-04

Requirement: Qualify the AWS model before relying on it for actions.

Planned acceptance: The approved gate retains 100% schema/currency/safety compliance and at least 95% exact intent-and-Constraint accuracy on its defined corpus; report unseen results separately.

ID: GR-05

Requirement: Provide traceable academic and individual evidence.

Planned acceptance: Confirmed owners, deadlines, weekly contribution records and final documentation/demo aligned to the actual DEPI requirements.

These are Must graduation requirements beyond the completed local MVP. Offline training and the seven-family development comparison have been performed on 210 training / 53 validation / zero unseen examples. The best selected validation result was 23/53 correct (43.40%) with macro-F1 0.4304; all data are synthetic/exposed. This fulfills exploratory experiment preparation, not independent model qualification or the final LLM comparison. CAP-02/03 final evaluation and AWS delivery remain open. The offline models do not enter the runtime or authorize Actions. Separate evaluation of context-dependent requests avoids claiming that isolated text classification measures full conversational execution.

Requirements traceability matrix

The matrix links every functional (FR), safety/quality (SQ) and graduation (GR) requirement to its origin, implementation responsibility and verification evidence. Requirement text and acceptance criteria remain defined above; the identifiers provide a two-way lookup from a requirement to its evidence and from an evidence group back to the requirements it covers. All 21 requirements are Must within the agreed scope.

Origin codes: **S1** = owner-approved local specification and domain model; **S2** = owner/tester feedback and Ticket 16 closeout; **S3** = approved graduation scope and ML backlog; **S4** = supplied DEPI documentation/submission guidelines. US identifiers refer to the user stories in this document; UC identifiers refer to the detailed design use cases.

Status codes: **Local** = implemented with recorded local evidence, not a universal correctness claim; **Limited** = locally available with stated capability/provider limits; **Partial** = some evidence exists but the full requirement remains open; **Open** = required future acceptance is not established. The matrix indexes existing records; it does not claim that the tests were rerun for this documentation update or that an informal study proves usability.

Functional traceability

Req. ID	Origin / story	Implementation	Verification / evidence	Status
FR-01	S1; US-01 / UC-01	M1: intent pipeline	V1: language, correction and context cases; unseen evaluation pending	Partial
FR-02	S1; US-01 / UC-01	M1, M2: discovery	V2: explicit constraints, stock and Exact Match/Alternative checks	Local
FR-03	S1, S2; US-02 / UC-02	M2: grounded advice	V2: read-only advice, references and fallback; R2 informal feedback	Limited
FR-04	S1; US-03 / UC-03	M3: navigation/Bridge	V3: observed destination, Spotlight and product-reference cases	Local
FR-05	S1; US-04 / UC-04-05	M4: cart/Undo	V4: exact variant/quantity, unaffected lines and Undo/refresh cases	Local
FR-06	S1; US-05 / UC-06	M4: Confirmation	V5: expiry, state binding, cancellation and single-use checks	Local
FR-07	S1; US-06 / UC-07	M3, M5: account guidance	V3, V6: login handback, newest-order Spotlight and sensitive exclusion	Local
FR-08	S1; US-07 / UC-08	M5: session lifecycle	V7: Stop, lease/takeover, stale/duplicate and refresh/reconnect checks	Local
FR-09	S1; US-08 / UC-09	M6: Panel/speech	V8: editable input and unavailable-speech fallback; typing remains available	Limited

Safety and quality traceability

Req. ID	Origin / story	Implementation	Verification / evidence	Status
SQ-01	S1; US-01	M1: intent boundary	V1: model-owned meaning; schema, capability and observed-target checks	Local
SQ-02	S1; US-06	M3: Bridge/privacy	V6: off-origin and Sensitive Field regressions; cloud checks still required	Local
SQ-03	S1; US-04-05	M1, M4: mutation authority	V4, V5, V7: stock/bounds, Confirmation and stale/duplicate rejection	Local
SQ-04	S1, S3	M2, M7: fallback/reporting	V2, V9; R1: preserved failures, usage unknowns and verified-card fallback	Local
SQ-05	S1, S3	M7: evaluation metrics	V9: phase timing and usage accounting; R1; cloud workload still open	Partial
SQ-06	S1, S4; US-06	M3, M6; data handling	V6, V8: sensitive exclusion; R2: privacy limits; cloud/data processes remain	Partial
SQ-07	S1; US-02	M2: read-only advice	V2: no executable advice fields; later mutations re-enter normal guards	Local

Graduation traceability

Req. ID	Origin / story	Implementation	Verification / evidence	Status
GR-01	S3; CAP-02	M7: dataset releases	R3: synthetic grouped releases exist; independent reviewed unseen set absent	Partial
GR-02	S3; CAP-03	M7: offline ML	V10; R3: 46 exploratory fits; independent final LLM comparison pending	Partial
GR-03	S3, S4; CAP-04-07	M8: AWS delivery	R4: design/cost constraints only; deployed journey and cloud acceptance pending	Open
GR-04	S3; CAP-05	M8: Bedrock qualification	R4: strict model gate specified; no qualified AWS model result recorded	Open
GR-05	S3, S4; CAP-01/08	Academic delivery process	R5: reports, assigned roles and owner approval/access record; upload/contributions incomplete	Partial

Implementation reference key

- **M1: agent/llm/intent_pipeline.py**, **agent/llm/intent.py**, **agent/discovery.py** and **agent/planner.py** interpret and validate supported shopping intent and plan bounded Actions.
- **M2: agent/catalogue.py** and **agent/advice.py** ground product results and read-only advice in catalogue evidence.
- **M3: agent/navigation.py**, **agent/product_reference.py**, **bridge/src/snapshot.ts**, **bridge/src/actions.ts** and **bridge/src/runtime.ts** provide destination/target checks, Snapshot privacy and browser execution.
- **M4: agent/cart.py**, **agent/confirmation.py**, **store/src/guarded-cart.ts** and **store/public/cart.js** implement cart authority, Confirmation and Undo.

- **M5: agent/sessions.py**, **agent/session_registry.py**, **agent/session_state.py**, **agent/session_stream.py** and **panel/src/event-stream.ts** coordinate task/session state and event delivery; the session facade delegates shopper commands and ActionResult/recovery processing to task modules.

- **M6: panel/src/panel.ts** and **agent/speech.py** provide the interaction panel and bounded speech service/fallback.

- **M7: eval/intent_benchmark.py**, **eval/baseline_pilot.py**, **eval/measurement.py**, **eval/milestone_report.py** and **eval/datasets/capstone-v1/** hold offline data/ML experiments and evaluation accounting. These models do not execute runtime Actions.

- **M8:** future AWS adapter, hosting, privacy/isolation and deployment controls are governed by CAP-04-07; this reference is an assigned work area, not implemented infrastructure.

Verification and evidence reference key

- **V1: agent/tests/test_intent_pipeline.py**, **agent/tests/test_semantic_boundary.py**, **agent/tests/test_product_context.py** and the real-model cases in **eval/milestone_cases.py**. Scripted tests verify execution contracts; only recorded real-model runs are language-quality evidence.

- **V2: agent/tests/test_catalogue.py**, **agent/tests/test_discovery.py**, **agent/tests/test_advice.py** and **store/tests/catalogue.test.ts**.

- **V3: eval/tests/test_ticket08_cases.py**, **agent/tests/test_real_task.py**, **bridge/tests/action-executor.test.ts** and product-reference cases in **agent/tests/test_product_context.py**.

- **V4: agent/tests/test_cart_task.py**, **store/tests/reversible-cart.test.ts** and **eval/tests/test_ticket11_cart.py**.

- **V5: agent/tests/test_confirmation.py**, **agent/tests/test_guarded_task.py**, **bridge/tests/guarded-mutation.test.ts** and **eval/tests/test_ticket10_guarded_browser.py**.

- **V6: bridge/tests/safety-review.test.ts**, **bridge/tests/snapshot-builder.test.ts** and **eval/tests/test_ticket09_safety.py**.

- **V7: agent/tests/test_sessions.py**, **agent/tests/test_recovery.py**, **panel/tests/event-stream.test.ts** and **eval/tests/test_ticket14_spa.py**.

- **V8: agent/tests/test_speech.py** and **panel/tests/panel-controller.test.ts**.

- **V9: eval/tests/test_measurement.py**, **eval/tests/test_milestone_report.py** and **eval/tests/test_usage_reconciliation.py**.

- **V10: eval/tests/test_baseline_pilot.py**, **eval/tests/test_group_allocation.py** and **eval/tests/test_model_comparison.py** verify offline experiment integrity, not an independent model-quality score.

- **R1: RESULTS.md** and **docs/ticket15-results.md** distinguish local release acceptance, historical live runs, measurements and limitations.

- **R2: docs/ticket16-closeout.md** records the actual informal feedback and owner-approved study adjustments.

- **R3: docs/cap03-model-comparison-results.md** and **notebooks/cap03-model-comparison.ipynb** document the synthetic/exposed 210-training / 53-validation / zero-unseen release.

- **R4: docs/cap04-credit-only-plan.md** and the CAP-04-07 backlog record cloud design/qualification/deployment requirements and outstanding decisions.
- **R5: docs/cap01-execution.md**, the supplied-guideline alignment register and the lecturer appendix record academic preparation and owner-confirmed approval/access; an actual official upload commit remains to be recorded.

The public paths above identify implementation and evidence locations, not proof that every requirement is fully accepted. Revisit affected rows when code, dataset release, model configuration, verification or approved scope changes; retain old results rather than silently replacing them.

Exclusions, assumptions and open questions

Excluded: arbitrary retailers, external store integrations, real transactions, multi-tenant SaaS and general production-readiness claims. The owner created an AWS Free-plan account; a 7 October screenshot showed $100 credits. The deployment constraint is zero out-of-pocket spending. Region/model eligibility, complete hosting/HTTPS costs, isolation, abuse limits and authorized provisioning remain CAP-04 decisions. Test fixtures and local accounts remain fictional; the current in-memory state is not assumed durable after cloud deployment.

The owner coordinates the overall submission and supplied the organization repository on 16 October, confirming both the invitation and lecturer approval were received. No additional approval wording/date or editable report template was supplied. Actual work is performed by the owner and coding assistant; nominal academic roles do not establish independent review. Dataset review/custody arrangements and numerical rubric weights remain unresolved. The published DEPI dates remain recorded; no formal extension is inferred from a relative submission date. Requirement changes should record their source, effect on implementation/evidence and explicit approval; passed tests do not silently redefine a requirement.

Export references and provenance

Source content dates are preserved. This PDF was exported on 16 October 2026; it is not an approval or official-upload record. Repository links below use the fixed development snapshot 503b7f6d0897f5a413d546952dd91c9c94d1dec4. They identify supporting evidence, not the official submission destination. External bibliography URLs are preserved as supplied and were not re-verified during export.

Editable source: docs/cap01-requirements.md
SHA-256: 948acba12e945c65322605b2dcb897f8c700155ccf2f946c6fae5eafccbe5e82

Linked sources

1. supplied content guidelines

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/depi-guideline-alignment.md

2. CAP-01 register

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/cap01-execution.md

3. Ticket 16 closeout

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/ticket16-closeout.md

4. design use cases

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/system-analysis-design.md#3-use-cases

5. RESULTS.md

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/RESULTS.md

6. seven-family development comparison

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/cap03-model-comparison-results.md

7. the organization repository

https://github.com/nhahub/NHA-5-145